

TASK 3

SQL, Python & Power BI Business Analysis

DIFFICULTY
Intermediate

DURATION
10 Days

DEADLINE
Saturday, 29 August 2026

OBJECTIVE

Perform an end-to-end business analysis on the Chinook database using SQL for data extraction, Python for analysis and visualization, and Power BI for dashboard reporting.

DATASET

Use the **Chinook SQLite Database**. It contains customers, invoices, tracks, albums, artists, genres and employees.

[Chinook SQLite Database - GitHub](#)

TOOLS

Python

Jupyter

SQLite

Pandas

Matplotlib

Power BI

1. SQL ANALYSIS

Write SQL queries to calculate and analyze the following:

- | | |
|----------------------------------|--|
| 1. Total Revenue | 11. Top-Selling Tracks |
| 2. Total Customers | 12. Revenue by Year |
| 3. Total Orders | 13. Revenue by Month |
| 4. Average Order Value | 14. Monthly / Yearly Sales Growth |
| 5. Average Spending per Customer | 15. Highest-Performing Country |
| 6. Top 10 Customers by Spending | 16. Highest-Performing Month |
| 7. Revenue by Country | 17. Highest-Performing Artist |
| 8. Customer Count by Country | 18. Customers with Highest Purchase Frequency |
| 9. Top-Selling Genres | 19. Revenue Contribution of Top Customers |
| 10. Top-Selling Artists | 20. High, Medium and Low-Value Customer Segments |

Required SQL concepts: SELECT, WHERE, GROUP BY, ORDER BY, HAVING, JOIN, SUM, AVG, COUNT, CASE, and at least one CTE or Subquery.

2. PYTHON ANALYSIS

Load SQL query results into Pandas DataFrames and create at least 6 meaningful visualizations:

- Revenue by Country
- Top 10 Customers
- Revenue by Genre
- Top Artists
- Monthly / Yearly Sales Trend
- Customer Distribution by Country

3. POWER BI DASHBOARD

Create one professional Power BI dashboard using the analyzed data.

KPI Cards

Total Revenue; Total Orders; Total Customers; Average Order Value

Required Visuals

At least 5 charts: Revenue by Country, Sales Trend, Top Customers, Revenue by Genre, Top Artists / Customer Distribution

Filters / Slicers

Country; Year; Genre

Quality

Clean layout, readable labels, consistent formatting and useful interaction

4. BUSINESS INSIGHTS

Write 5-7 short business insights. Each insight must include:

- Finding
- Supporting value / number
- Recommended action

Example

USA generated the highest revenue of \$X. The company should prioritize retention and targeted campaigns in this market.

DELIVERABLES

01 analysis.ipynb	05 Power BI dashboard (.pbix)
02 analysis.sql	06 5-7 business insights
03 Minimum 20 SQL queries	07 README.md
04 Minimum 6 Python visualizations	

SUBMISSION RULE

All submitted SQL queries, Python code, calculations and dashboard elements must be understandable and editable by the intern during evaluation.